

March 26, 2025

Amazon Sales Analysis

1 Introduction

[Amazon sales product dataset](#) contains product information, including pricing, discounts, ratings, reviews, and user interactions. The following questions are designed to analyze different aspects of the dataset using statistical methods:

- Does the discounted price significantly impact product ratings?
 - Are product categories and high/low ratings independent?
 - Is there a significant difference in ratings between high-discount and lowdiscount products?
 - Do more expensive products receive higher ratings on average?
 - Does the distribution of rating counts follow a normal distribution?
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2 Data Exploration & Preprocessing

2.1 Dataset Overview

- The dataset consists of **1465 rows** and **15 columns**.

#	Column Name	Non-Null Count	Data Type
1	product_name	1465	object
2	category	1465	object
3	discounted_price	1465	object
4	actual_price	1465	object
5	discount_percentage	1465	object
6	rating	1465	object
7	rating_count	1463	object
8	about_product	1465	object
9	user_id	1465	object
10	user_name	1465	object
11	review_id	1465	object
12	review_title	1465	object
13	review_content	1465	object
14	img_link	1465	object
15	product_link	1465	object

2.2 Handling NaN Data and Data Cleaning

- For the Kolmogorov-Smirnov Test, “rating_test” has 2 NaN values that were dropped.
- All the features are in object format and for numerical calculations should be transformed to numeric type.
- one of the entities had “|” (pipe) for “rating”. dropped.

2.3 Distributions

- To divide the data into groups for statistical analysis, box plots were used to identify quantiles and select appropriate bins [1]

3 Statistical Analysis

3.1 Does the discounted price significantly impact product ratings?

- Method: Spearman correlation
- Analysis:
 1. Spearman Correlation: 0.076
 2. P-value: 0.004
- Result:

The correlation is almost negligible ($\rho = 0.076$), meaning that X and Y have little relationship. Statistical Meaning: The p-value (0.004) suggests this weak correlation is unlikely due to random chance. Statistical Significance Does NOT Mean Strong Relationship. A small p-value only means there is evidence that ρ is different from zero. However, it doesn't mean the correlation is strong or meaningful. Spearman Measures Monotonicity, Not Linearity. Even if Spearman's ρ is low, there could be non-monotonic (but still structured) relationships in the data.

3.2 Are product categories and high/low ratings independent?

- Method: chi2_contingency
- Analysis:
 - The number of rows(categories) is too big(211), and it causes big DoF(Degree of Freedom) So we reduce the categories to the major categories.
 - The expected Frequencies for some categories are smaller than 5, so the test may not be valid. So we remove it for the test to be reliable.
 - Chi-Square Statistic: 32.492
 - P-value: 0.00000
 - Degrees of Freedom: 3
 - Expected Frequencies: $\begin{bmatrix} 288.21619767 & 164.78380233 \\ 334.66163349 & 191.33836651 \\ 284.39876458 & 162.60123542 \end{bmatrix}$ $\begin{bmatrix} 19.72340426 & 11.27659574 \end{bmatrix}$
- Result:

The analysis shows that categorizing the products is significant to the rating. Since the p-value is very small ($p < 0.05$), we reject the null hypothesis. This means there is a significant

association between the variables being tested. In other words, the categorical variables are not independent.

3.3 Is there a significant difference in ratings between high-discount and low-discount products?

- Method: T-test
- Analysis:
 - Dividing data based on discount_percentage to 2 groups, 'high' and 'low'.

discount_level	mean_value	varaince_value	group_size
High	18,000	4.064926	0.096302
Low	23,500	4.130282	0.071451

- t-Statistic: 14.760
- P-value: 0.00000
- Result:

With such a large dof, the t-test behaves almost like a z-test (normal distribution). The result of p_value is saying: "The difference is so significant that it is almost impossible to occur by chance." Since the data from both groups are similar and normal, the assumptions for the t-test (such as normality) are met.[2]. Although the p-value is very small (close to 0), the difference in means might not be practically significant. The t-test just shows that the observed difference is statistically significant.

3.4 Do more expensive products receive higher ratings on average?

- Method: ANOVA(F-statistic)
- Analysis:
 - Price distribution is skewed based on the box plot[3]; using quantiles is a good approach to dividing products into price groups.
 - Between-group variance: 0.088
 - Within-group variance: 0.085
 - F-statistic: 1.038
 - critical value: 3.0019129829411924
 - P-value: 0.35433684306160806
- Result:

The f-statistic is smaller than the critical F-value. so we fail to reject the null hypothesis, indicating that the groups are not significantly different. The P_value also confirms the result (it's much bigger than, for example, 0.05)

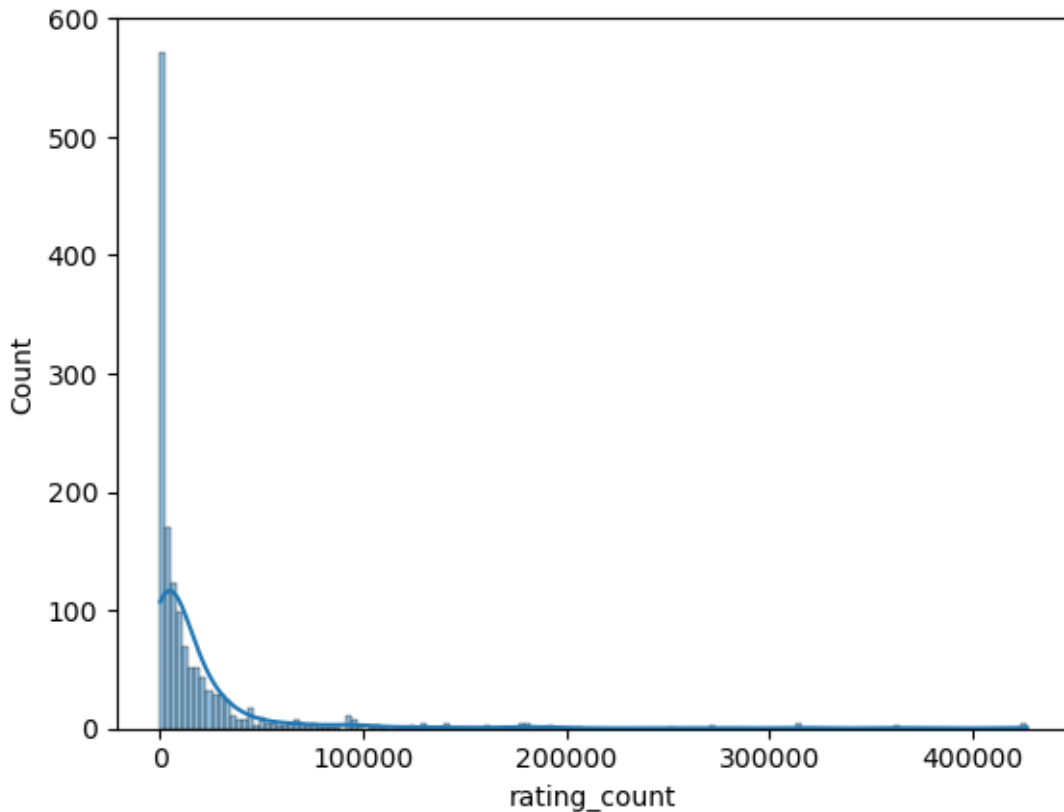
3.5 Does the distribution of rating counts follow a normal distribution?

- Method: Kolmogorov-Smirnov Test
- Analysis:

- Kolmogorov-Smirnov Statistic (D_n): 0.33384
- Kolmogorov-Smirnov Test Statistic using Python package: 0.3345
- P-value: 0.0000

- Result:

A Kolmogorov-Smirnov (KS) test statistic of 0.3345 means that the maximum difference between the empirical cumulative distribution function (ECDF) of your rating_count data and the cumulative distribution function (CDF) of a normal distribution is 33.45%. So, the data is not well-matched to a normal distribution. And since P_value is 0, it rejects the null hypothesis (data is not normal).



4 Discussion & Conclusion

This report analyzed various statistical relationships between product prices, discounts, categories, and ratings to assess their significance in determining consumer behavior. Below are the key findings:

1. Discounted Price vs. Ratings The Spearman correlation analysis revealed a negligible relationship ($\rho = 0.076$) between discounted price and product ratings. While the correlation was statistically significant ($p = 0.004$), it is not practically meaningful.
2. Product Categories vs. Ratings The Chi-Square test showed a significant association between

product categories and ratings ($\chi^2 = 32.492$, $p < 0.05$). This suggests that product category influences ratings, rejecting the assumption of independence.

3. Discount Level vs. Ratings

The t-test indicated a statistically significant difference in ratings between high-discount and low-discount products ($t = 14.76$, $p < 0.0001$). However, while statistically significant, the practical difference in means was small.

4. Price vs. Ratings

ANOVA results ($F = 1.038$, $p = 0.354$) showed no significant difference in ratings across different price groups. This suggests that higher-priced products do not necessarily receive higher ratings.

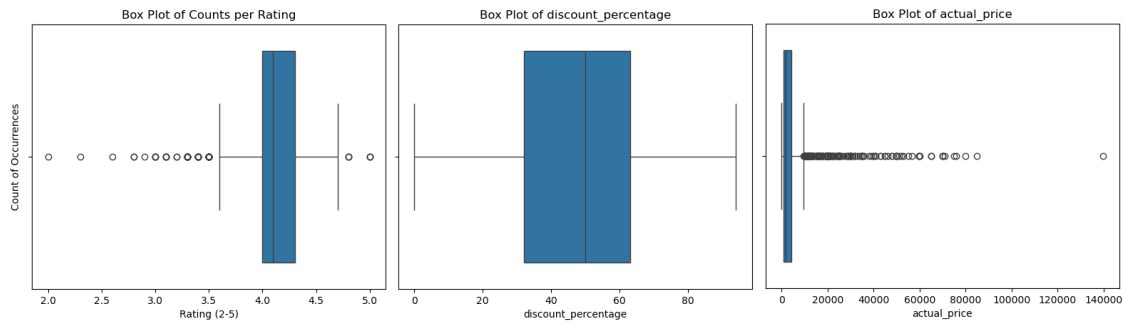
5. Distribution of Rating Counts

The Kolmogorov-Smirnov test ($D = 0.3345$, $p < 0.0001$) confirmed that rating counts do not follow a normal distribution. The large deviation from normality suggests that parametric tests relying on normal assumptions may not be appropriate for this variable.

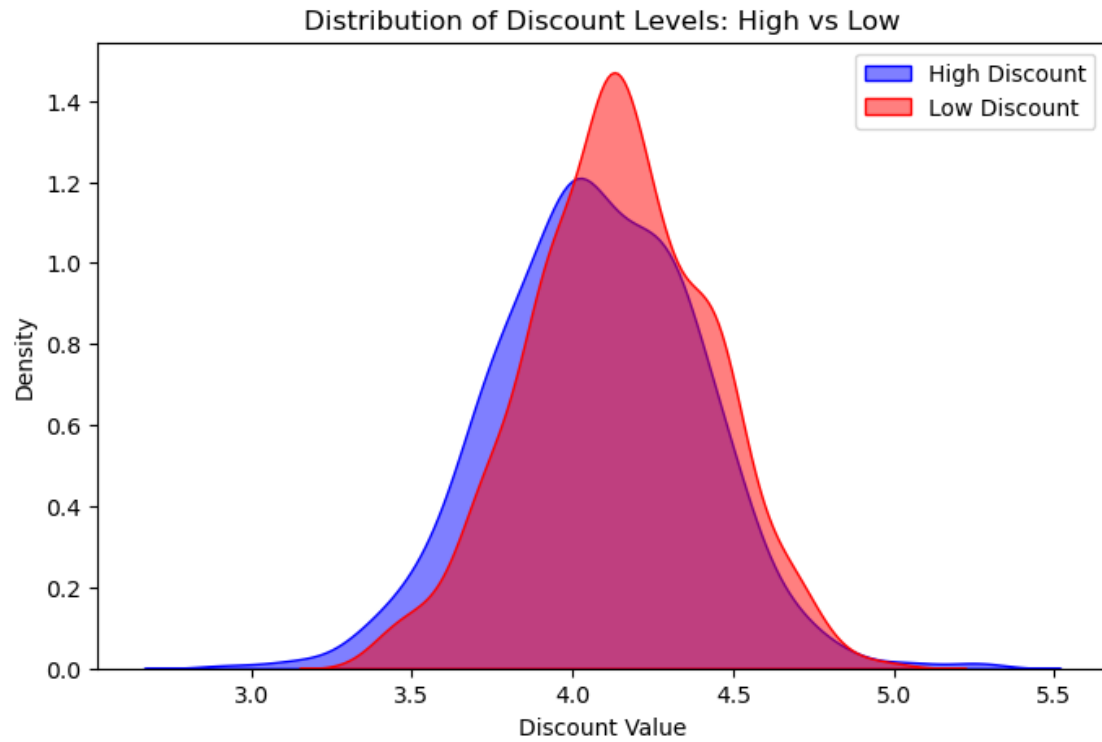
- Final Insights While statistical significance was found in multiple cases, practical significance varied. The relationship between discounts and ratings, while statistically significant, was weak. Product categories showed a stronger association with ratings, but price had little influence. The findings suggest that consumer ratings are influenced more by categorical product attributes than by pricing or discounts alone. Future analyses could incorporate more complex models to capture non-linear relationships and additional factors influencing consumer behavior.

5 Appendix

- [1] : distributions explained by box plot



- [2] : Distribution of discount levels



- [3] : Price distribution

